

Transformations

US Common Core – High School Geometry

Domain: Congruence (G-CO.A) | Cluster: Experiment with transformations in the plane

1 Overview

Definition

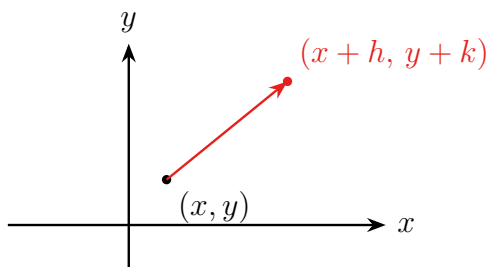
A **transformation** moves every point of a figure according to a rule. The three rigid transformations covered here are:

- (a) **Translation** – shifting a point.
- (b) **Rotation** – rotation of a point about the origin.
- (c) **Reflection** – reflection of a point through a line.

2 Translation

Rule

$$(x, y) \mapsto (x + h, y + k)$$



Example 1

Rule: $(x, y) \mapsto (x - 8, y + 2)$. Find the image of $A(2, 1)$.

$$A' = (2 - 8, 1 + 2) = (-6, 3)$$

Example 2: Translating a square

Square $ABCD$ with $A(2, 1)$, $B(5, 1)$, $C(5, 5)$, $D(2, 5)$, under the rule $(x, y) \mapsto (x - 7, y + 7)$:

$$A'(-5, 8), \quad B'(-2, 8), \quad C'(-2, 12), \quad D'(-5, 12)$$

Example 3: Translation preserves differences

$A(2, 1)$ and $B(4, 1)$ under the rule $(x, y) \mapsto (x + 1, y + 1)$:

$$A'(3, 2), \quad B'(5, 2)$$

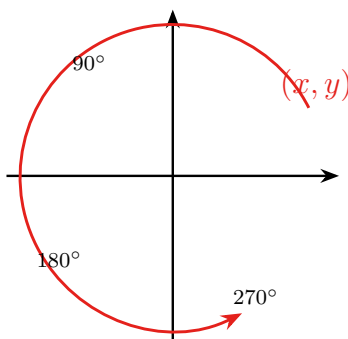
Notice $B - A = (2, 0)$ and also $B' - A' = (2, 0)$ – a translation shifts every point by the same amount, so distances and differences between points never change.

3 Rotation About the Origin

Rotation Rules

$$90^\circ : (x, y) \mapsto (-y, x) \quad 180^\circ : (x, y) \mapsto (-x, -y) \quad 270^\circ : (x, y) \mapsto (y, -x)$$

Note: 180° is two 90° rotations composed, and 270° is three 90° rotations composed – applying $(x, y) \mapsto (-y, x)$ repeatedly derives all three rules.



Example 4

$$(a) \quad (4, 2) \xrightarrow{90^\circ} (-2, 4)$$

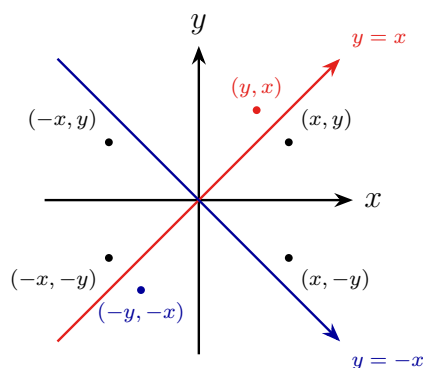
$$(b) \quad (-7, 5) \xrightarrow{90^\circ} (-5, -7) \xrightarrow{90^\circ} (7, -5) \xrightarrow{90^\circ} (5, 7) \quad [\text{a full } 270^\circ]$$

$$(c) \quad (-1, -3) \xrightarrow{90^\circ} (3, -1) \xrightarrow{90^\circ} (1, 3) \quad [\text{a full } 180^\circ]$$

4 Reflection

Reflection Rules

Over the x -axis: $(x, y) \mapsto (x, -y)$
Over the y -axis: $(x, y) \mapsto (-x, y)$
Over the line $y = x$: $(x, y) \mapsto (y, x)$
Over the line $y = -x$: $(x, y) \mapsto (-y, -x)$



Example 5: Reflecting a triangle J, K, L

	Over x -axis	Over y -axis
$J(2, -4)$	$J'(2, 4)$	$J(5, 3) \rightarrow J'(-5, 3)$
$K(3, 7)$	$K'(3, -7)$	$K(1, -2) \rightarrow K'(-1, -2)$
$L(6, -1)$	$L'(6, 1)$	$L(3, 4) \rightarrow L'(-3, 4)$
	Over $y = x$	Over $y = -x$
$J(2, -4)$	$J'(-4, 2)$	$J(5, 3) \rightarrow J'(-3, -5)$
$K(7, 3)$	$K'(3, 7)$	$K(1, -2) \rightarrow K'(2, -1)$
$L(-1, 6)$	$L'(6, -1)$	$L(-4, 3) \rightarrow L'(-3, 4)$

5 Reflection Over a General Vertical or Horizontal Line

Rules

Over the vertical line $x = k$: $(x, y) \mapsto (2k - x, y)$
Over the horizontal line $y = k$: $(x, y) \mapsto (x, 2k - y)$

Example 6

Reflect $(-9, -8)$ over the line $x = 1$:

$$(2(1) - (-9), -8) = (11, -8)$$

6 Composite Transformations

Idea

Transformations can be **composed** – applying one after another. Order matters: reflecting-then-translating is generally different from translating-then-reflecting.

Translation after Reflection

Reflect (x, y) over the x -axis, then translate by (h, k) :

$$(x, y) \xrightarrow{R} (x, -y) \xrightarrow{T} (x + h, -y + k)$$

For $A(1, 2)$ reflected over the x -axis then translated by $(x - 7, y + 7)$:

$$A(1, 2) \xrightarrow{R} (1, -2) \xrightarrow{T} (-6, 5)$$

Reflection after Translation

Translate (x, y) by (h, k) , then reflect over the line $y = -x$:

$$(x, y) \xrightarrow{T} (x + h, y + k) \xrightarrow{R(y=-x)} (-(y + k), -(x + h))$$

For $A(1, 2)$ translated by $(x - 7, y + 7)$ then reflected over $y = -x$:

$$A(1, 2) \xrightarrow{T} (-6, 9) \xrightarrow{R(y=-x)} (-9, 6)$$

Translation after Rotation, and Rotation after Translation

Using $A(5, 2)$ and 180° rotation with translation rule $(x, y) \mapsto (x + 1, y - 1)$:

$$\text{Rotate, then translate: } A(5, 2) \xrightarrow{180^\circ} (-5, -2) \xrightarrow{T} (-4, -3)$$

$$\text{Translate, then rotate: } A(5, 2) \xrightarrow{T} (6, 1) \xrightarrow{180^\circ} (-6, -1)$$

Notice the two different orders give two different final points – **order matters**.

Rotation, then reflection over a line, then rotation again

A four-step composition on $A(1, 2)$: rotate 90° , reflect over the line $x = 1$, rotate 270° , then reflect over the line $y = 3$:

$$(1, 2) \xrightarrow{90^\circ} (-2, 1) \xrightarrow{R(x=1)} (4, 1) \xrightarrow{270^\circ} (1, -4) \xrightarrow{R(y=3)} (1, 10)$$

Note: The source notes include further exploratory pages composing two reflections (e.g. over $y = x$ and $y = -x$) to see when the result matches a single rotation. The general pattern – composing two reflections over intersecting lines is equivalent to a single rotation about their intersection point, by twice the angle between the lines – is a well-known enrichment result, though the source’s own worked trace of it is inconclusive (one branch is marked with an $\times$ where the shortcut did not match), so it has not been reproduced as a stated rule here.